PRANAY DAS

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PROFESSIONAL EXPERIENCE

Eli Lilly and Company **August 2025 - Present**
Software Engineer *Remote*
- Designed a slot reservation service for radiopharmaceutical dose ordering used by 10K+ clinical technicians, using idempotent requests and optimistic concurrency control to lock each slot on the first confirmed booking, preventing race-condition double bookings by design.
- Built a patient context service for adverse event classification that retrieves live medical history and redacts PII before model input, serving context in ~3 ms.
- Developed a citation verification component that checks model explanations against retrieved historical cases, contributing to ~70% of adverse event cases bypassing manual classification review.
- Built Python transformers mapping canonical case data to MuleSoft and Veeva formats, and reviewed teammates' PRs for the downstream pipeline stages that run after an authorized reviewer approves a submitted case change.

Google Summer of Code **May 2024 - August 2024**
Software Engineer *Remote*
- Built and published **scorecode**, a Python package that retrieves OpenSSF Scorecard supply-chain security metrics; adopted as a ScanCode.io dependency, with ~2K monthly PyPI downloads.
- Integrated Scorecard results into ScanCode.io through reusable Django models and CycloneDX SBOM export support, with pytest coverage for public and private repositories.
- Collaborated with open-source maintainers through iterative code reviews, refining API design, class naming conventions, documentation, and code quality to meet project contribution standards.

Instoried **August 2019 - May 2023**
Software Developer (Founding Engineer) *Bengaluru, Karnataka, India*
- Led a team of 5–6 backend, frontend, and DevOps engineers in an agile, feature-ownership model; hired ~5 engineers and drove technical design, code reviews, and mentoring alongside the CTO.
- Built the backend from scratch (Python, FastAPI, PostgreSQL) for an AI writing assistant recommending tone, emotion, and concision improvements, scaling to 100K+ registered users.
- Built an OAuth 2.0 authorization service and developer portal that let ~20 B2B customers integrate the recommendation APIs directly into their own products.
- Designed a Git-like text versioning system using snapshots and incremental deltas, sending only changed chunks instead of the full document (up to ~10 MB) on every save, cutting payloads to a few KB and letting any lost chunk be restored from the last snapshot.
- Resolved serving failures under 3× campaign traffic spikes by adding Redis caching (content-hashed keys, TTL, LFU eviction), cutting repeat recommendation latency from 1–2 s to ~1 ms, plus Kubernetes horizontal pod autoscaling.
- Built a real-time sales dashboard from scratch, fed by an incremental, checkpoint-based ETL pipeline that synced onboarding and campaign data without loading the primary database.

KaptureCRM **January 2019 - May 2019**
Software Developer *Bengaluru, Karnataka, India*
- Designed a bidirectional sync backend connecting Shopify, HubSpot, NetSuite, Snowflake, and SQL databases to the CRM through a canonical object model, so each source needed only its own mapper rather than point-to-point logic, supporting 50+ onboarded clients.
- Matched change capture to each source type (webhooks for SaaS platforms, WAL-based CDC for SQL databases, and offset-based polling workers for legacy systems), propagating changes in both directions in near real time with eventual consistency.

Aptus Data Labs **June 2017 - December 2018**
Software Developer *Bengaluru, KA, India*
- Built 5 REST API connectors, mostly sources such as QuickBooks and Zendesk, for an early-stage data migration platform that moved user-authorized data, limited to granted scopes, into destinations such as PostgreSQL.
- Contributed to a global data schema that let the middleware read heterogeneous sources in a unified way, replacing source-to-destination-specific transformation logic.
- Built Java-based Azure and JSON data-source plugins for Yellowfin BI, letting users pull data from these sources and visualize it directly in the BI tool.

SKILLS

Programming Languages: Python, C/C++, Java
Web Frameworks & APIs: FastAPI, Flask, Django, Node.js, REST APIs, gRPC, GraphQL, OpenAPI, Swagger, pytest
Cloud & DevOps: AWS, S3, Lambda, Kubernetes, Git, CI/CD, Terraform, Jenkins, Google Cloud Platform, BigQuery, Docker, GitHub Actions
Databases: PostgreSQL, Supabase, MySQL, MongoDB, Redis, Cassandra, Elasticsearch, DynamoDB
Architecture & Infrastructure: Temporal, Apache Kafka, Modular Monolith, Microservices
AI Tools: Claude Code, CoPilot

PROJECTS & OUTSIDE EXPERIENCE

Compliance Checker
Freelance Software Developer (Novo Nordisk) - Pharmaceuticals/Healthcare *Remote*
- Designed and developed the complete backend using Python and FastAPI to extract information from clinical PDFs, check documents against client-defined compliance rules, flag violations, and apply corrections.
- Replaced an EC2-based PDF processing service with parallel AWS Lambda functions, splitting 400-page documents into smaller tasks of 20–30 pages that could be processed in 6–10 seconds each.
- Built a reliable processing pipeline using DynamoDB to combine extraction results, with automatic retries and dead-letter queues to handle failed tasks and flag unresolved issues for developer review.

KYC Portal
Freelance Software Developer (Natixis Investments) - FinTech *Remote*
- Designed and developed a KYC/AML screening platform using Python, FastAPI, and PostgreSQL, replacing spreadsheet-based workflows and reducing screening and review turnaround from approximately one week to 8 hours.
- Built a Kafka-based pipeline to collect financial data from Moody's, Refinitiv World-Check, and internal banking APIs in parallel, combining results into a common data format for analysts to review.
- Implemented real-time screening updates using Server-Sent Events (SSE) and SSO-based access, allowing analysts to track progress, review reports, add comments, and collaborate across teams.

EDUCATION

Purdue University - West Lafayette **June 2023 - May 2025**
Master's, Computer Science

Sambalpur University **August 2013 - April 2017**
Bachelor's, Computer Science